

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled Trial Comparing the Efficacy and Safety of Polatuzumab Vedotin in Combination With Rituximab and CHP (R-CHP) Versus Rituximab and CHOP (R-CHOP) in Previously Untreated Patients With Diffuse Large B-Cell Lymphoma **Short Title/Acronym:** POLARIX **Protocol Number:** G039942 **NCT Number:** NCT03274492 **Posting ID:** 187939002460679087 **Sponsor Name:** Hoffmann-La Roche / Genentech **Company:** Roche **Trial Status:** Active, not currently recruiting (ACTIVE_NOT_RECRUITING) **Investigational Product:** Polatuzumab Vedotin (Polivy) + R-CHP (Rituximab, Cyclophosphamide, Doxorubicin, Prednisone) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To compare the efficacy, safety, and pharmacokinetics of polatuzumab vedotin in combination with R-CHP versus standard-of-care R-CHOP in participants with previously untreated diffuse large B-cell lymphoma (DLBCL). **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Complete Response (CR) rate, and long-term tolerability, including the incidence and severity of peripheral neuropathy and neutropenia.

2.2 Background & Rationale Initial treatment for DLBCL traditionally consisted of rituximab plus cyclophosphamide, doxorubicin, vincristine, and prednisone (R-CHOP). While effective, a significant proportion of patients still relapse or are refractory to frontline therapy. Polatuzumab vedotin is a CD79b-directed antibody-drug conjugate that targets cancer cells directly. This study aims to determine if substituting vincristine with polatuzumab vedotin in the standard R-CHOP regimen (creating Pola-R-CHP) improves Progression-Free Survival (PFS) and overall disease control in previously untreated intermediate- or high-risk DLBCL patients.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator, Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (1:1 ratio)

3.2 Planned Enrollment & Duration Target \$\$\$: ~1,000 participants (Target recruitment) **Number of Sites:** Multicenter (Global)
Estimated Study Start: November 2017 **Estimated Primary Completion:** 30-Apr-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 to ≤ 80 years. 2. Previously untreated participants with CD20-positive DLBCL (including DLBCL NOS, double-hit/triple-hit lymphoma, and high-grade B-cell lymphoma). 3. International Prognostic Index (IPI) score of 2 to 5. 4. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0, 1, or 2. 5. Life expectancy ≥ 12 months. 6. Left ventricular ejection fraction (LVEF) $\geq 50\%$ and adequate hematologic/organ function.

4.2 Exclusion Criteria 1. History of severe allergic or anaphylactic reactions to humanized or murine monoclonal antibodies. 2. Current Grade > 1 peripheral neuropathy by clinical examination or history of demyelinating Charcot-Marie-Tooth disease. 3. History of indolent lymphoma, CNS lymphoma, follicular lymphoma grade 3B, grey-zone lymphoma, primary mediastinal (thymic) large B-cell lymphoma, Burkitt lymphoma, or prior organ transplantation. 4. Clinically significant cardiovascular disease, or known active viral infections (e.g., HIV, Hepatitis B/C).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive either Polatuzumab vedotin + R-CHP (Rituximab, Cyclophosphamide [ATC Code: L01AA01; Trade names: Cytosan, Endoxana], Doxorubicin, Prednisone) + vincristine placebo OR R-CHOP + polatuzumab vedotin placebo every 21 days for up to 6 cycles. **Route of Administration:** Intravenous infusion / Oral (prednisone) **Duration of Treatment:** 6 treatment cycles (18 weeks), followed by a long-term follow-up phase.

5.2 Concomitant & Prohibited Therapy Permitted: Primary prophylaxis with granulocyte colony-stimulating factor (G-CSF) is required for all patients during the first 6 cycles. Intrathecal chemotherapy for CNS prophylaxis is permitted. **Prohibited:** High-dose chemotherapy for CNS prophylaxis, any investigational therapy within 28 days prior to cycle 1, and prior receipt of anthracyclines.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures	:---	:---
:---	Screening	Day -28 to 0 (2-4 weeks)	Informed	

Consent, Medical History, LVEF evaluation (ECHO/MUGA), Baseline PET/CT | | **Treatment Cycles 1-6** | Every 21 days | Randomization, IV/Oral Drug Administration, Safety Labs, G-CSF prophylaxis | | **Interim Assessment** | After Cycle 4 | Interim efficacy assessment via PET/CT scan | | **End of Treatment** | 6 to 8 weeks post-Tx | Final treatment PET/CT scan, comprehensive safety assessment | | **Follow-Up** | Q3M / Annually | Long-term tracking for survival, PFS, and delayed AEs |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS), defined as the time from randomization to the first occurrence of disease progression, relapse, or death from any cause.

Measurement Method: Investigator-assessed based on clinical evaluations and PET/CT scan imaging.

7.2 Secondary & Exploratory Endpoints 1. Event-Free Survival (EFS) and Overall Survival (OS). 2. Complete Response (CR) rate at the end of treatment measured by PET/CT. 3. Safety endpoints, highlighting grade ≥ 3 neutropenia and peripheral neuropathy incidence.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of AEs at every study visit, with specific monitoring for cytopenias, febrile neutropenia, and peripheral neuropathy via clinical examination. **Serious Adverse Events (SAEs):** Must be reported within 24 hours to the sponsor. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to evaluate unblinded safety and efficacy data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary efficacy analyses; Safety Analysis Set for patients receiving at least one dose of the study drug. **Sample Size Justification:** $\sim 1,000$ participants recruited globally to adequately power the study to detect a statistically significant PFS benefit of Pola-R-CHP over standard R-CHOP. **Handling of Missing Data:** Standard oncology methods for right-censoring patients at the last date of evaluable disease assessment for time-to-event endpoints (PFS/OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring of participating centers to assure protocol compliance. **Audit Trail:** Secure EDC (Electronic Data Capture) systems, eCRF locking, and rigorous central imaging review (independent review facility) for PET/CT scans.

11. Study Identifiers & Governance

Primary ID: G039942 **Registry Number:** NCT03274492 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** POLARIX **Official Regulatory Title:** A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled Trial Comparing the Efficacy and Safety of Polatuzumab Vedotin in Combination With Rituximab and CHP (R-CHP) Versus Rituximab and CHOP (R-CHOP) in Previously Untreated Patients With Diffuse Large B-Cell Lymphoma

12. Clinical Framework (DLBCL Specific)

Primary Condition: Diffuse Large B-Cell Lymphoma (DLBCL) **Disease Semantic Type:** Neoplastic Process **MeSH Code:** D016403 **SNOMED ID:** 109969005 **Diagnosis Threshold:** Histologically confirmed CD20+ DLBCL with IPI score of 2 to 5 **Medical Methodology:** Polatuzumab vedotin + R-CHP targeted immunochemotherapy vs standard R-CHOP **Optimization Target:** Progression-Free Survival (PFS) and prevention of relapse

13. Study Procedures & Methodology

Management Pathway: Standard frontline systemic immunochemotherapy + G-CSF prophylaxis **Education Protocol (HCP):** LVEF safety monitoring & peripheral neuropathy clinical grading standards **Education Protocol (Patient):** Informed consent regarding chemotherapy side effects (alopecia, neutropenia, neuropathy) **Quality Audit Mechanism:** Centralized independent review of PET/CT scans for unbiased response evaluation

14. Observation & Follow-Up Schedule

Total Duration: Up to ~9 years total per protocol timelines (Start 2017 to Est. Comp 2026) **Onsite Visit Cadence:** Every 21 days for 6 cycles, followed by specific interval visits post-treatment **Remote Monitoring Frequency:** Standard safety check-ins as clinically indicated. **Checklist Review Interval:** Restaging every 3 months during the initial post-treatment follow-up period

Role	Name	Signature	Date		:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					